WORKSHEET 5.5

# Absolute Extrema on a Closed Interval

Unit 5: Analysing Functions

## Key ideas

- **Candidates test.** On a closed interval, the absolute extrema occur at critical points or at endpoints.
- Evaluate $f$ at every critical point in the interval and at both endpoints. The largest value is the absolute maximum; the smallest is the absolute minimum.
- The list of values *is* the justification. Show it.
- On an open interval with a single critical point, a sign change of $f'$ over the whole interval gives an absolute extremum.

## Worked examples

### Example 1 · The candidates test

Find the absolute extrema of $f(x) = x^3 - 3x^2 + 1$ on $[-1, 4]$.

**Solution.** $f'(x) = 3x(x-2)$. $f(-1) = -3$, $f(0) = 1$, $f(2) = -3$, $f(4) = 17$.

**Maximum 17 (at $x = 4$); minimum $-3$ (at $x = -1$ and $x = 2$).**

### Example 2 · A trigonometric function

Find the absolute extrema of $f(x) = \sin x + \dfrac{x}{2}$ on $[0, 2\pi]$.

**Solution.** $f'(x) = \cos x + \dfrac{1}{2} = 0$ at $\dfrac{2\pi}{3}$ and $\dfrac{4\pi}{3}$. $f(0) = 0$, $f\left(\dfrac{2\pi}{3}\right) \approx 1.913$, $f\left(\dfrac{4\pi}{3}\right) \approx 1.228$, $f(2\pi) = \pi$.

**Maximum $\pi$ (at $x = 2\pi$); minimum 0 (at $x = 0$).**

### Example 3 · An open interval

Find the absolute minimum of $f(x) = x^2 + \dfrac{54}{x}$ for $x > 0$.

**Solution.** $f'(x) = 2x - \dfrac{54}{x^2}$ is negative for $0 < x < 3$ and positive for $x > 3$.

**The absolute minimum is $f(3) = 27$.**



AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

## Practice

### CORE

1. Find the absolute extrema of $f(x) = x^2 - 2x$ on $[0, 3]$.

2. Find the absolute extrema of $f(x) = 4 - x^2$ on $[-1, 2]$.

3. Find the absolute extrema of $f(x) = x^3$ on $[-2, 1]$.

4. Why must the endpoints be included in the candidates test?

### EXAM LEVEL

5. Find the absolute extrema of $f(x) = 2x^3 - 9x^2 + 12x$ on $[0, 3]$.

6. Find the absolute extrema of $f(x) = xe^{-x/2}$ on $[0, 6]$.

7. Find the absolute extrema of $f(x) = x^2 - 8 \ln x$ on $[1, e]$.



AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

8. Find the absolute extrema of $f(x) = |x - 2|$ on $[0, 5]$.

**Question 9 · Free-response practice**

Let $f(x) = 3x^4 - 4x^3 - 12x^2 + 2$.

(a) Find the critical points of $f$.

(b) Find the absolute maximum value of $f$ on $[-2, 3]$. Justify your answer.

(c) Find the absolute minimum value of $f$ on $[-2, 3]$.



AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

## Worked solutions

1. $f(0) = 0$, $f(1) = -1$, $f(3) = 3$: maximum 3, minimum $-1$.
2. $f(-1) = 3$, $f(0) = 4$, $f(2) = 0$: maximum 4, minimum 0.
3. $f(-2) = -8$, $f(0) = 0$, $f(1) = 1$: maximum 1, minimum $-8$.
4. An extremum on a closed interval can occur at an endpoint, where the derivative need not be zero.
5. $f'(x) = 6(x-1)(x-2)$. $f(0) = 0$, $f(1) = 5$, $f(2) = 4$, $f(3) = 9$: maximum 9, minimum 0.
6. $f'(x) = e^{-x/2}\left(1 - \dfrac{x}{2}\right) = 0$ at $x = 2$. $f(0) = 0$, $f(2) = \dfrac{2}{e}$, $f(6) = \dfrac{6}{e^3}$: maximum $\dfrac{2}{e}$, minimum 0.
7. $f'(x) = 2x - \dfrac{8}{x} = 0$ at $x = 2$. $f(1) = 1$, $f(2) = 4 - 8\ln 2 \approx -1.545$, $f(e) = e^2 - 8 \approx -0.611$: maximum 1, minimum $4 - 8\ln 2$.
8. The only critical point is $x = 2$, where $f'$ does not exist. $f(0) = 2$, $f(2) = 0$, $f(5) = 3$: maximum 3, minimum 0.
9. (a) $f'(x) = 12x^3 - 12x^2 - 24x = 12x(x-2)(x+1)$: $x = -1$, $x = 0$ and $x = 2$.
   (b) $f(-2) = 34$, $f(-1) = -3$, $f(0) = 2$, $f(2) = -30$, $f(3) = 29$. The absolute maximum is 34, at the endpoint $x = -2$.
   (c) From the same list, the absolute minimum is $-30$, at $x = 2$.



AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.